AERIS-H743

PCB component placement plan — pre-routing

Fabrication-oriented placement plan for the 6-layer, 5-inch-class quadcopter PCB. Zones and priorities are derived from the confirmed mechanical spec (160mm hole-to-hole diagonal, 46x46mm hub, 20mm arms) and the established schematic architecture. Anything not yet specified in the project (exact battery current, motor/ESC current draw, U1 switching frequency, firmware axis convention) is marked OPEN/NEEDS VERIFICATION rather than assumed.

1. Per-component placement table

Ref	Component	Zone	Orientation/relation	Reason
J1	XT30 connector	Power entry	Board edge, cable exit pointing outward	First point of contact — needs external cable access
Q1	AO3407A P-FET	Power entry	In line between J1 and F1, within 5mm of both	Reverse-polarity protection must be first in the chain, no detours
R1	10k gate pulldown	Power entry	Directly at Q1 gate pin	Minimize gate trace length
F1	Fuse 2A PPTC	Power entry	In line, within 5mm of Q1 and R26	Series element, keep the high-current path straight
D1	SMF26A TVS	Power entry	Directly across the F1-R26 junction to nearest GND via	Needs shortest possible path to ground to clamp effectively
R26	Shunt 20mOhm	Power entry	In line, Kelvin-sense traces to U10 kept short and symmetric	Current-sense accuracy depends on clean, undisturbed sense trace routing
C1, C2	Buck input caps 10uF	Power conversion	Within 3mm of U1 VIN pin	High-frequency input decoupling, distance directly increases loop inductance
U1	AP63205WU buck	Power conversion	Oriented so SW node/inductor loop is compact and away from IMU zone	Switching node is the primary EMI source on the board
L1	Buck inductor	Power conversion	Immediately adjacent to U1 SW pin, shortest possible SW trace	SW node trace should be as short and wide as practical — high dv/dt
C3	Buck output cap 22uF	Power conversion	Within 3mm of L1 output / U1 FB divider	Output ripple filtering, close to the node it filters
R2, R3	FB divider	Power conversion	Tight to U1 FB pin, away from SW node/inductor	FB is a sensitive high-impedance node — noise pickup here destabilizes regulation
C4	Bootstrap cap 100nF	Power conversion	Directly between U1 BOOT and SW pins	Bootstrap caps must be immediately adjacent to their pins by design
U9a	AP2112K-3.3 digital LDO	Power conversion	Between buck output and MCU zone	Shortens the +3V3 digital distribution path
U9b	AP7343-33W5-7 sensor LDO	Power conversion	Between buck output and IMU/sensor zone, physically separated from U9a	Keeps sensor rail regulation independent and close to what it feeds
C5a-d	LDO in/out caps 1uF	Power conversion	Directly at each LDO IN/OUT pin	Standard LDO stability requirement — distance affects transient response
U10	INA226	Power entry	Adjacent to R26, Kelvin sense traces direct	Sense accuracy depends on undisturbed direct connection to shunt
C7	INA226 decoupling 100nF	Power entry	At U10 VCC pin	Standard IC decoupling
J2	USB-C receptacle	USB zone	Board edge, connector shell grounded to nearest edge copper	Needs external cable access
U2	USBLC6-2SC6 ESD	USB zone	Between J2 and the D+/D- trace path, within 3mm of J2 pins	ESD protection must intercept transients before the trace, at the connector
R4, R5	USB series 22R	USB zone	In line between U2 and MCU USB pins	Standard USB series termination placement
R6, R7	CC pulldowns 5.1k	USB zone	At J2 CC1/CC2 pins	Local to the pins they set
D2	PMEG3020EH power-OR diode	Power entry	Between J2 VBUS and the +5V bus junction	Sits on the power path, not a signal path

Ref	Component	Zone	Orientation/relation	Reason
U3	STM32H743VIT6	MCU zone	Hub center, orientation set so QSPI/SDMMC face memory zone and USB pins face USB zone	Central position minimizes average trace length to every other zone — see MCU orientation note below
C-VDD x5	100nF VDD decoupling	MCU zone	One capacitor within 2mm of each of the 5 VDD pins, on the same layer where possible	High-frequency decoupling effectiveness drops sharply with distance — this is the single most important passive-placement rule on the board
C-bulk	4.7uF VDD bulk	MCU zone	Within 5mm of the VDD pin cluster	Bulk capacitance can sit slightly further than the 100nF caps
C-VCAP1, C-VCAP2	2.2uF VCAP caps	MCU zone	Directly at VCAP_1 and VCAP_2 pins, shortest possible traces, low-inductance via to GND	These supply the MCU core regulator — ST explicitly calls out low-ESR/short-trace requirement in AN4938
C6, C18	VDDA decoupling 1uF/100nF	MCU zone	At VDDA pin, via FB1 ferrite	Analog supply — isolate from digital switching noise on the same rail
C13, C33	VREF+ decoupling 1uF/100nF	MCU zone	At VREF+ pin	Same reasoning as VDDA
FB1	Ferrite bead	MCU zone	In line between +3V3 and VDDA pin	Filters digital supply noise before it reaches the analog supply
SW1	Reset button	MCU zone (edge-accessible)	Near board edge if possible for physical access during bring-up	You will press this a lot during firmware development
C-nrst	NRST filter 100nF	MCU zone	At NRST pin	Standard filter placement
SW2	BOOT0 button	MCU zone (edge-accessible)	Near SWD header for convenient access during flashing	Used together with reset during bootloader entry
R8	BOOT0 pulldown 10k	MCU zone	At BOOT0 pin	Standard
X1	25MHz HSE crystal	MCU zone	As close as physically possible to OSC_IN/OSC_OUT pins, symmetric trace lengths, guard-ringed with GND pour, no other signals routed nearby	Crystal traces are highly sensitive to parasitic capacitance and coupling — this is a hard placement rule, not a preference
C-hse1, C-hse2	HSE load caps	MCU zone	Directly at crystal pins, symmetric placement	Asymmetric placement unbalances the oscillator loop
X2	32.768kHz LSE crystal	MCU zone	Same rules as X1, physically separated from X1 and from any switching supply	Low-frequency crystal is even more sensitive to coupling than HSE
C-lse1, C-lse2	LSE load caps	MCU zone	Directly at crystal pins	Same as HSE
J3	SWD header	SWD zone (edge-accessible)	Board edge, oriented for a programmer clip or pogo fixture	Needs physical access throughout bring-up and later debugging

Ref	Component	Zone	Orientation/relation	Reason
U4	ICM-42688-P IMU	IMU zone	Hub center, offset toward the side away from power conversion, one axis aligned to arm 1 — OPEN/NEEDS VERIFICATION: confirm firmware axis convention before finalizing orientation	Most vibration- and noise-sensitive component on the board — see IMU placement note above
C23	IMU decoupling 100nF	IMU zone	Directly at U4 VDD pin	Standard, but especially important for a sensitive analog/MEMS device
R14	IMU CS pullup 10k	IMU zone	At U4 CS pin	Standard
U6	BME280	Other-sensor zone	MCU-adjacent, can tolerate closer proximity to digital signals than the IMU	Environmental sensor, not vibration-critical
C24	BME280 decoupling 100nF	Other-sensor zone	At U6 VDD pin	Standard
U7	DS3231MZ RTC	Other-sensor zone	MCU-adjacent, away from crystal X1/X2 to avoid frequency interference	RTC has its own timing crystal-equivalent internally — keep clear of the MCU crystals
C25, C26	RTC decoupling	Other-sensor zone	At U7 VCC and VBAT pins	Standard
BT1	CR1220 holder	Other-sensor zone	Accessible for battery replacement — consider board edge or a cutout	This is a user-serviceable part, unlike everything else on the board
R12, R13	I2C1 pullups 4.7k	Other-sensor zone	Near the physical middle of the I2C1 bus run (U6/U7/U10), not at any single endpoint	Standard I2C pullup placement practice
U5	W25Q128JVSIQ QSPI flash	Memory zone	MCU-adjacent, shortest practical QSPI trace length, length-matched CLK/IO0-3	QSPI is a source-synchronous high-speed bus — trace length mismatch directly costs timing margin
C-flash	Flash decoupling 100nF	Memory zone	At U5 VCC pin	Standard
R-qspi2, R-qspi3	QSPI IO2/IO3 pullups	Memory zone	Near U5, on the QSPI_IO2/IO3 nets	Standard
J4	microSD socket	Memory zone (edge-accessible)	Board edge or accessible cutout for card insertion/removal, SDMMC traces kept short from MCU	Physical card access is a hard mechanical requirement, not optional
C21, C22	microSD decoupling	Memory zone	At J4 VDD pin	Standard
R9-R13	SDMMC pullups 47k	Memory zone	Near J4/SDMMC bus run	Standard

Ref	Component	Zone	Orientation/relation	Reason
U13	MCP2515 (if used) or native FDCAN path	Comms zone	Near CAN connector J5, away from IMU zone	OPEN/NEEDS VERIFICATION: confirm final CAN architecture (native FDCAN vs earlier MCP2515 plan) before finalizing this row
U8	SN65HVD230 CAN transceiver	Comms zone	Directly adjacent to J5 connector	Transceiver should sit as close as possible to the physical bus connector
C-can	CAN decoupling 100nF	Comms zone	At U8 VCC pin	Standard
R17	CAN termination 120R (DNP)	Comms zone	Across J5 CANH/CANL, right at the connector	Termination belongs at the physical bus endpoint, not upstream
J5	CANH/CANL header	Comms zone (edge-accessible)	Board edge	External bus connector
U11	MAX3485 RS485 transceiver	Comms zone	Directly adjacent to J6 connector	Same reasoning as CAN transceiver
C-485	RS485 decoupling 100nF	Comms zone	At U11 VCC pin	Standard
R18	RS485 termination 120R (DNP)	Comms zone	Across J6 A/B, at the connector	Same as CAN termination
J6	RS485 A/B header	Comms zone (edge-accessible)	Board edge	External bus connector
J7	GPS header	Comms zone (top edge)	Top edge of hub, per original mechanical plan	Keeps GPS antenna cable routing away from motor/ESC arms — reduces EMI exposure
Ref	**Component**	**Zone**	**Orientation/relation**	**Reason**
J10-J13	ESC signal pads x4	Arm tips	At each arm tip, near the mounting hole	Motors and ESCs physically live here — no flexibility on this one
R19-R22	ESC signal pulldowns 10k	Arm tips	At each J10-J13 pad	Local to the pin they protect
J8, J9	Expansion headers	Hub edge, wherever space remains	Accessible edge, non-critical	Lowest placement priority — fill remaining space last
LED1-3	Status LEDs	Hub edge, visible when assembled	Should be visible without disassembly	User-facing indicator, needs line of sight
SW3	User button	Hub edge, physically accessible	Reachable without disassembly	Same reasoning as LEDs

2. Forbidden and problematic placements

Component/zone	Keep away from	Minimum practical separation	Consequence if ignored
IMU (U4)	U1 buck converter and L1 inductor	Maximum practical distance within the 46mm hub — realistically 15-20mm; OPEN/NEEDS VERIFICATION exact figure depends on U1 switching frequency (not yet specified)	Switching noise couples into IMU readings, causing bias drift and noisy attitude estimates — directly affects flight stability
IMU (U4)	Any high-current trace or pour, including the SW node copper	No shared copper layer directly beneath the IMU footprint	Magnetic field coupling from current-carrying copper corrupts gyroscope readings
X1, X2 crystals	Any digital switching signal (QSPI, SDMMC, USB, PWM)	No parallel routing within 3mm, no crossing on adjacent layers without a solid ground plane between them	Coupled noise causes clock jitter, which cascades into timing errors on every peripheral that depends on the system clock
USB differential pair	U1 SW node and buck inductor	Route on the opposite side of the MCU from the buck if possible	Switching noise injected into the USB pair causes enumeration failures or data errors, especially at USB high-speed rates
R26 shunt / INA226 Kelvin sense traces	Any other current-carrying trace running parallel	Dedicated, undisturbed sense trace pair, no shared copper with the power path itself	Sense accuracy degrades — you get an inaccurate current reading, defeating the purpose of the current-monitoring circuit
J5/J6 (CAN, RS485 connectors)	IMU zone and crystal zone	Opposite side of the hub where mechanically possible	External cable-borne noise/ESD events have a shorter, more direct coupling path to sensitive circuitry
J1 (battery connector) and Q1/F1/D1	Board edge clearance	Standard fab-house minimum edge clearance for connectors, typically 0.5-1mm depending on your fab — confirm with JLCPCB before finalizing	Connector courtyard violates DFM rules, risking fabrication rejection or assembly issues
J4 (microSD socket)	Any component that would block the card ejection path	Physical clearance for the card and your fingers, not just the socket footprint	A card you cannot physically remove without desoldering something is a real, embarrassing failure mode

3. Routing-priority table (before you route anything)

Category	Nets	Priority	Requirement
Differential pair	USB_DP / USB_DM	Critical	Length-matched pair, controlled impedance per your stackup (see section 5), routed together with consistent spacing
High-speed single-ended, matched group	QSPI_CLK, QSPI_IO0-3	Critical	Length-matched within the group, short and direct, minimal via count
High-speed single-ended, matched group	SDMMC_CK, SDMMC_CMD, SDMMC_D0-3	Critical	Same reasoning as QSPI
Clock	HSE (X1), LSE (X2) crystal traces	Critical	Shortest possible, symmetric, guard-ringed with ground pour, isolated from all other signals
High-current power	+BATT path (J1 through Q1, F1, R26, to U1 VIN)	Critical	Wide traces or copper pour, sized for actual current draw — OPEN/NEEDS VERIFICATION: exact width depends on your final battery/current decision
Switching power	U1 SW node to L1	Critical	Short, wide, minimal loop area — this is your primary EMI source
Sensitive analog	IMU SPI bus (CS, SCK, MOSI, MISO)	High	Short, direct, away from any switching or high-current trace on any layer
Sensitive analog	INA226 Kelvin sense pair	High	Dedicated undisturbed pair, see forbidden-placement table
Bus, shared	I2C1_SCL, I2C1_SDA	Medium	Standard routing, pullups placed mid-bus
Bus, differential-ish	CANH, CANL	Medium	Matched length, routed as a pair even though not strictly controlled-impedance
Bus, differential-ish	RS485 A, B	Medium	Same as CAN
UART	GPS TX/RX, RS485 TX/RX (MCU side)	Low	Standard single-ended routing
GPIO/control	ESC1-4 PWM, RGB LED, status LED, user button	Low	No special requirements beyond keeping clear of the critical categories above
Power distribution	+5V, +3V3, +3V3_SENSOR	High	Prefer plane/pour distribution over traces where the layer stack allows
Ground	GND	Critical	Solid plane, not routed as traces — see stackup strategy

4. 6-layer stackup strategy (JLC06161H-3313)

This is guidance based on the general structure of this stackup class (signal-ground-signal-power-ground-signal or similar 6-layer arrangements are standard for this JLCPCB offering) — OPEN/NEEDS VERIFICATION: pull the exact per-layer assignment and dielectric heights from JLCPCB's current stackup documentation for JLC06161H-3313 before finalizing impedance-controlled trace widths, since those numbers directly depend on it and I have not independently verified the current spec sheet in this conversation.

Layer	Typical assignment for this class	Placement implication
L1 (top)	Components + primary signals (USB diff pair, QSPI, SDMMC, SPI1)	Keep high-speed signals here where possible, referenced to L2 ground
L2	Ground plane	Never split, even under the buck converter — this is the return path for every signal on L1
L3	Power plane(s) — split pour for +5V/+3V3/+3V3_SENSOR	Keep the +3V3_SENSOR pour physically separated from +3V3 digital, moated, single connection point back to the LDO
L4	Ground plane	Second ground reference, improves return path for L5 signals
L5	Secondary signals (I2C1, UART, GPIO, ESC PWM, CAN/RS485)	Lower-speed signals that do not need L1-quality reference
L6 (bottom)	Components + remaining signals	Keep the buck converter switching loop confined to as few layers as possible to minimize radiated loop area